

Dongho Lee

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Research Interests

Keywords : Symplectic Geometry and Dynamics, Celestial Mechanics

My primary research focuses on understanding dynamical systems through the framework of symplectic geometry. For now, I am particularly interested in bifurcation phenomena of Hamiltonian systems. Using tools from Floer theory, I investigate the geometric and topological mechanisms underlying these behaviors. I am also interested in connecting these theoretical insights to practical applications in systems celestial mechanics such as the restricted three-body problem, and also other related areas.

Education

Ph.D. in Mathematics

Department of Mathematical Science, Seoul National University (SNU) Mar 2018 – Feb 2025

Bachelor of Science in Mathematics

Bachelor of Economics in Economics, Minor in Physics

College of Liberal Studies, SNU

Mar 2013 – Feb 2018

Graduated with Honors

Research Experience

Postdoctoral Researcher

Center for Quantum Structures in Modules and Spaces (QSMS), SNU May 2025 – Present

Research Institute of Mathematics, SNU

Mar 2025 – Apr 2025

Postdoctoral Supervisor: Prof. Cheol-hyun Cho

Graduate Student Researcher

Department of Mathematical Sciences, SNU

Mar 2018 – Feb 2025

Ph.D Advisor: Prof. Otto van Koert

Publications

1. Conley-Zehnder Indices of Spatial Rotating Kepler Problem

arXiv 2506.14325

Journal of Topology and Analysis, 27 Feb 2026

2. Global Hypersurfaces of Section and the Spatial Kepler Problem

Advisor: Prof. Otto van Koert

Ph.D. Thesis, Dec 2024

3. Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces

arXiv 2401.20667, with Sunghae Cho

Archiv der Mathematik, Vol. 123, 31 Jul 2024

Talks and Presentations

Sildes and posters are available at my homepage.

Invited Talks

1. **Local Floer Chain Complex across Hamiltonian Bifurcation**
Geometry Space Surrey, University of Surrey, Guildford, UK 10 June 2026
2. **Conley–Zehnder Indices and Bifurcations of the Spatial Rotating Kepler Problem**
Oyster Dynamics Seminar, Dalian University of Technology, Dalian, China 19 May 2026
3. **Bifurcation of Periodic Orbits in Hamiltonian Systems**
G-Lamp Mathematics Seminar, Soongsil University, Seoul, Korea 22 Jan 2026
4. **Generic Bifurcation of Hamiltonian Orbits**
Symplectic Learning Seminar, SNU, Seoul, Korea 12 Dec 2025
5. **Hamiltonian Dynamics and Three-body Problem**
Rookie's Pitch, SNU, Seoul, Korea 14 Oct 2025
6. **Three-body Problem and Related Problems**
QSMS 2025 Summer Workshop, Goseong, Korea 21 Aug 2025
7. **Periodic Orbits of Rotating Kepler Problem**
Billiards and Quantitative Symplectic Geometry, Heidelberg, Germany 16 Jul 2025
8. **Closed orbits of the spatial rotating Kepler problem**
2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea 25 Apr 2025

Poster Talks

1. **Periodic Orbits and Symplectic Invariants in the Rotating Kepler Problem**
Celestial Mechanics & N-body Dynamics, Tokyo, Japan 16 Nov 2025

Conference Participation

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| International Conference on Celestial Mechanics
SUSTech, ShenZhen, China | 20 – 24 Apr 2026 |
| New Developments in Symplectic Geometry
Seoul National University, Seoul, Korea | 25 – 28 Nov 2025 |
| East Asian Symplectic Conference 2025
Hokkaido University, Sapporo, Japan | 25 – 29 Sep 2025 |
| 2025 Symplectic Retreat
Tongyeong, Korea | 23 – 26 Jan 2025 |
| From Hamiltonian Dynamics to Symplectic Topology and Beyond
Institut Henri Poincaré, Paris, France | 3 – 7 Jun 2024 |
| Topology of 4-manifolds and Related Topics
Jeju, Korea | 21 – 27 Jan 2024 |
| East Asian Symplectic Conference 2023
Jeju, Korea | 29 Oct – 4 Nov 2023 |

QSMS Workshop on Symplectic Geometry and Related Topics Jeju, Korea	5 – 10 Feb 2023
Workshop on Symplectic Dynamics Instituto Superior Técnico, Lisboa, Portugal	27 Jun – 1 Jul 2022
Symplectic Geometry and Beyond (Part I) Jeju, Korea	7 – 10 Feb 2022
Fukaya 60 Geometry and Everything Kyoto University, Kyoto, Japan	17 – 22 Feb 2019

Personal Affiliations

Korean Mathematical Society , Korea	2025 – Present
QSMS-BK21 Symplectic Seminar , Korea	2021 – Present

Teaching Experience

Teaching and Course Assistant , SNU	Mar 2018 – Feb 2025
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Major Courses

Algebraic Topology (Graduate Course)	Fall 2019
Introduction to Topology	Spring 2023, Spring 2020
Introduction to Differential Geometry	Spring, Fall 2021
Selected Topics Seminar (College of Liberal Studies)	Spring 2023

General Education Courses

Head Teaching Assistant for Introductory Mathematics Courses	Spring 2020
Engineering Mathematics	Fall 2024, Fall, Summer, Spring 2023, Fall 2020
Calculus for Life Science	Fall 2022, Spring 2021
Fundamentals and Applications of Mathematics	Spring 2022
Calculus Practice	Fall, Spring 2018
Calculus	Fall, Spring 2018

Awards and Honors

Merit-based Scholarship , Department of Mathematical Sciences, SNU	2018
National Scholarship , Korean Student Aid Foundation	2015
Merit-based Scholarship , College of Liberal Studies, SNU	2013

References

Prof. Otto van Koert Department of Mathematical Sciences, SNU okoert@snu.ac.kr Relationship: Ph.D. Advisor	Prof. Cheol-hyun Cho Department of Mathematics, POSTECH chocheol@postech.ac.kr Relationship: Postdoctoral Supervisor
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